

macOS Digital Forensics



Artefact Analysis & Command Line Essentials

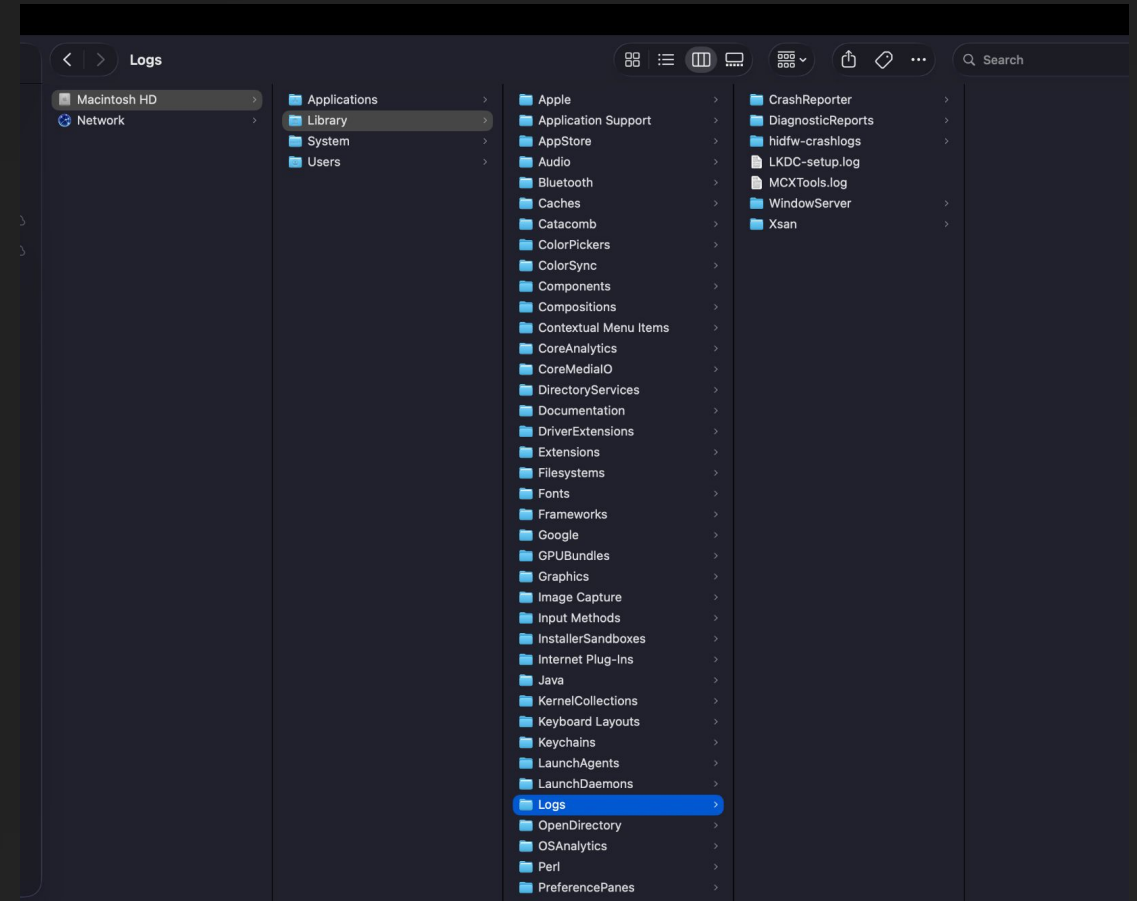
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1. macOS System Logs Overview

Unified Logging System (logd)

- Logs record user actions, application events, and system operations.
- **Main Types:**
 - System Logs
 - Security Logs
 - Application Logs

```
log show --info --last 1h
log show --predicate 'eventMessage contains "login"'
--last 24h
console
```



Important Log Locations

Key Directories

Understanding where logs reside is crucial for manual extraction and analysis.

- **/var/log/system.log**: Traditional system log file.
- **/private/var/log/**: System-wide logs requiring elevated privileges.
- **/Library/Logs/**: Application and system service logs.
- **~/Library/Logs/**: User-specific application logs (CrashReporter, etc.).

Extraction Commands



```
# View system log cat /var/log/system.log
# List log directories ls /Library/Logs/
ls ~/Library/Logs/
# Collect logs to zip sudo log collect --output
logs.zip
```

Artefacts from Logs

What to Look For

Logs provide a timeline of critical user and system activities.

- **Authentication:** Login / logout timestamps & failed attempts.
- **Privilege Escalation:** sudo usage commands.
- **Execution:** App launch & crash logs.
- **File System:** Activity traces and modification events.

Analysis Commands



```
# Check recent logins last
# Check who is logged in who
# Find sudo usage events sudo log show --info
--predicate 'eventMessage contains "sudo"' --last 5m
```

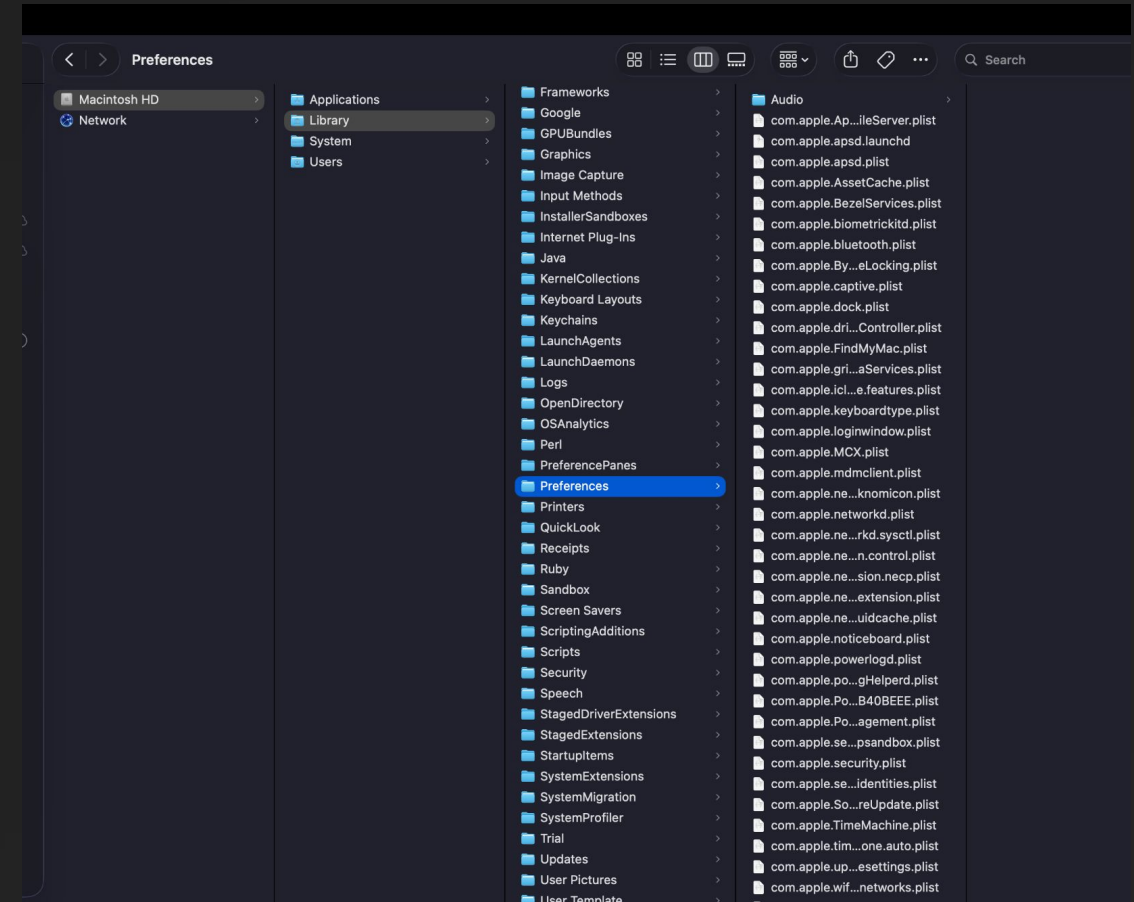
2. User & Application Preferences

.plist Files

Configuration and preference data is stored in Property List (plist) files.

- **User Level:** ~/Library/Preferences/
- **System Level:** /Library/Preferences/

```
plutil -p ~/Library/Preferences/com.apple.finder.plist  
ls ~/Library/Preferences/
```



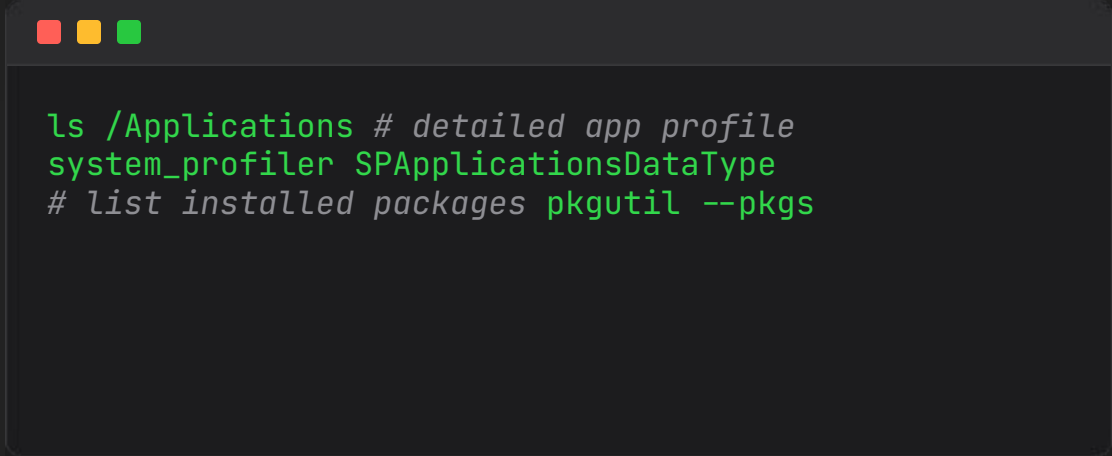
Installed Applications

Locations & Receipts

Tracking installed software helps identify authorized tools vs. potential malware.

- **/Applications**: System-wide apps.
- **~/Applications**: User-specific apps.
- **/Library/Receipts/**: History of package installations and updates.

Enumeration Commands



```
ls /Applications # detailed app profile  
system_profiler SPApplicationsDataType  
# list installed packages pkgutil --pkgs
```

Forensic Importance

Why analyze apps?

Detect Anomalies

Identifies unauthorized or suspicious applications installed on the system.

History & Behavior

Reveals installation timelines and user customization behavior.

Software Links

Links specific users to the usage of specific software tools.

Search Command



```
# Find all application bundles  
mdfind "kMDItemContentType =  
'com.apple.application-bundle'"
```

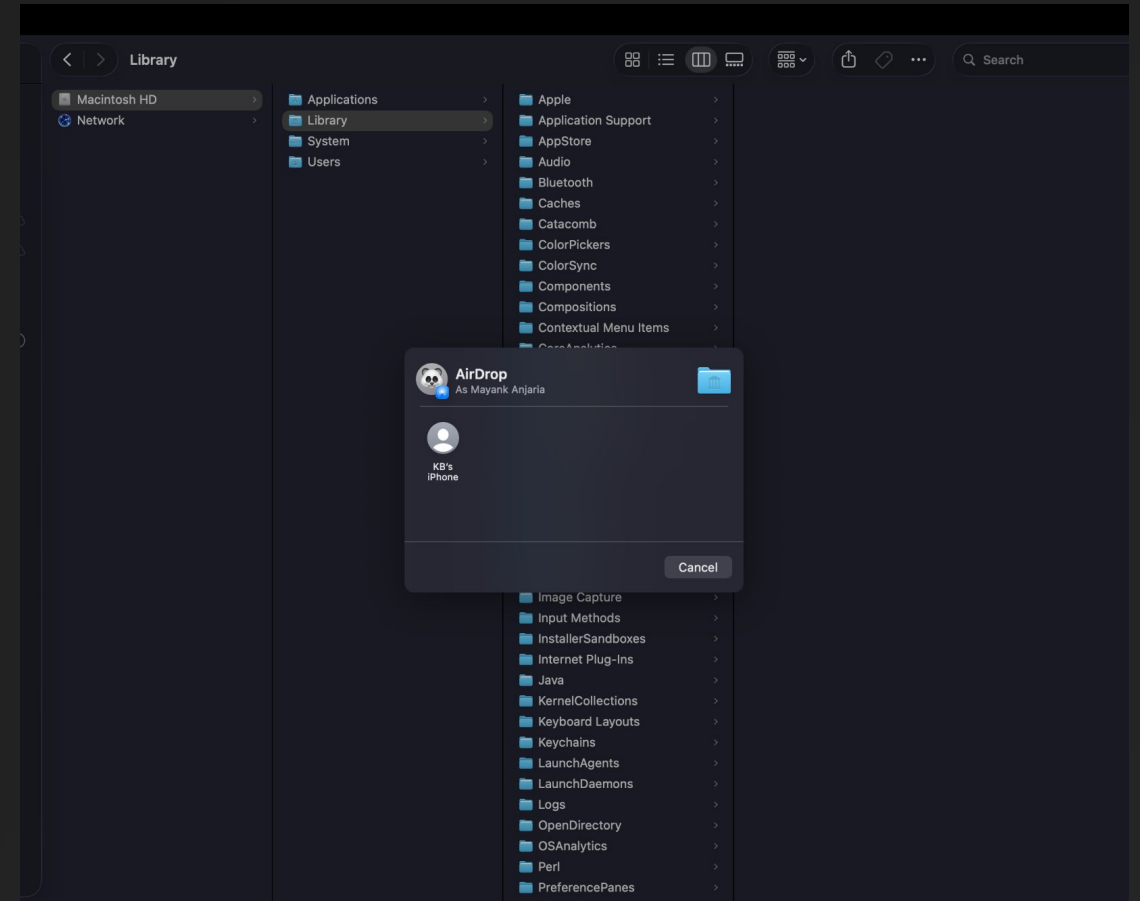
3. Attached Devices & Shared Locations

Device Tracking

macOS maintains detailed records of hardware connections.

- USB Mass Storage Devices
- External Hard Drives (HDD/SSD)
- Mobile Devices (iOS/Android)

```
diskutil list
system_profiler SPUSBDataType
ioreg -p IOUSB
```



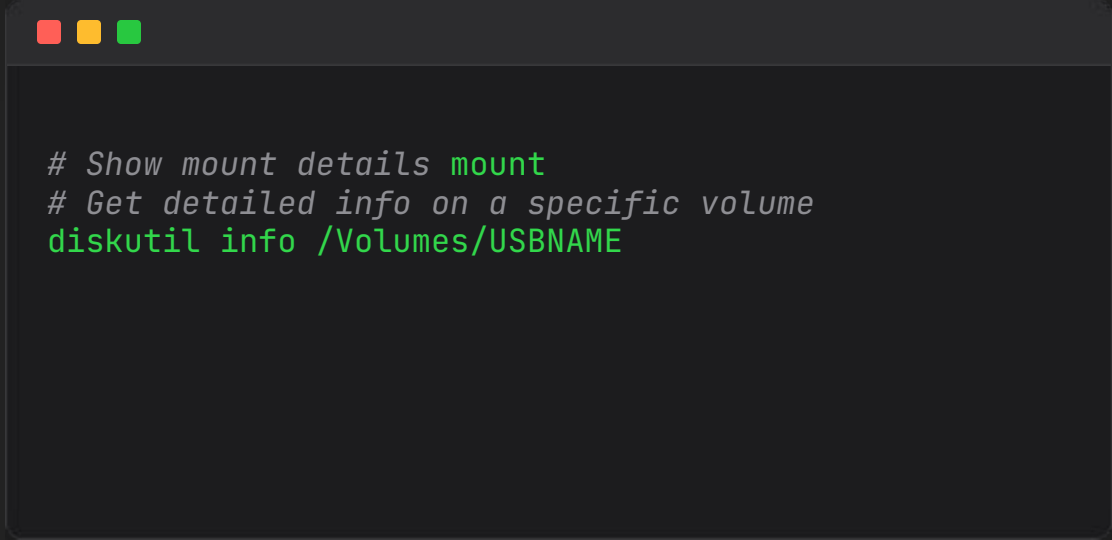
Artefacts from External Devices

Critical Information

When an external device is connected, the system captures:

- **Volume Name:** The label given to the drive.
- **Mount Point:** Where it was accessed in the file tree (e.g., /Volumes/MyDrive).
- **File System Type:** APFS, HFS+, ExFAT, etc.

Inspection Commands



```
# Show mount details mount
# Get detailed info on a specific volume
diskutil info /Volumes/USBNAME
```

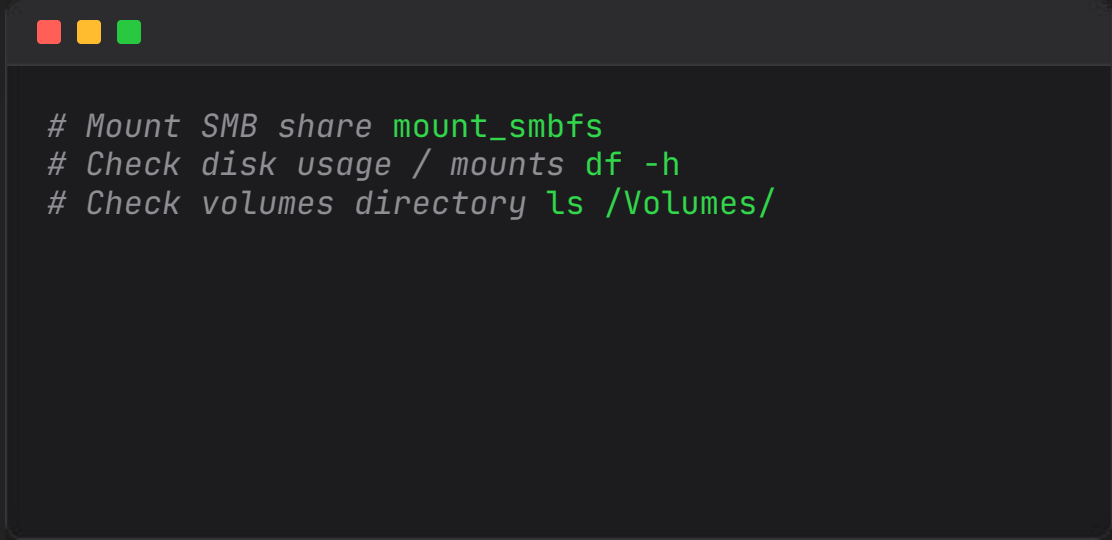
Shared Locations

Network Shares

Artefacts related to network file access can indicate data exfiltration or lateral movement.

- **SMB:** Server Message Block (Windows shares).
- **AFP:** Apple Filing Protocol (Legacy Mac shares).
- **NFS:** Network File System (Unix/Linux).

Network Mount Commands



```
# Mount SMB share mount_smbfs  
# Check disk usage / mounts df -h  
# Check volumes directory ls /Volumes/
```

4. Recently Accessed Documents

User Activity Traces

Recent items help reconstruct what files the user was working on.

- **Finder Recent Items:** Stored in user library.
- **Spotlight Metadata:** Indexes file attributes and access times.

```
# List recent items ls -lt ~/Documents  
# View metadata for a file mdls filename
```



Programs & Location Artefacts

LaunchServices & Quarantine

These databases track which applications open specific file types and where files originated (e.g., downloaded from the web).

- **LaunchServices:** Maps documents to apps.
- **QuarantineEvents:** Logs downloaded file sources (URL, Date).

Database Commands



```
ls ~/Library/Preferences/com.apple.LaunchServices*  
# Query quarantine database  
sqlite3 ~/Library/Preferences/com.apple.  
LaunchServices.QuarantineEventsV2
```

Forensic Significance on macOS

Building the Narrative

Timeline

Helps build a chronological timeline of user activity.

Patterns

Shows file access patterns and frequency.

Usage History

Tracks application usage history and preferences.

Timeline Commands



```
# List files sorted by time ls -lt ~/Downloads  
# Detailed file status (MAC times) stat filename
```

Thank You